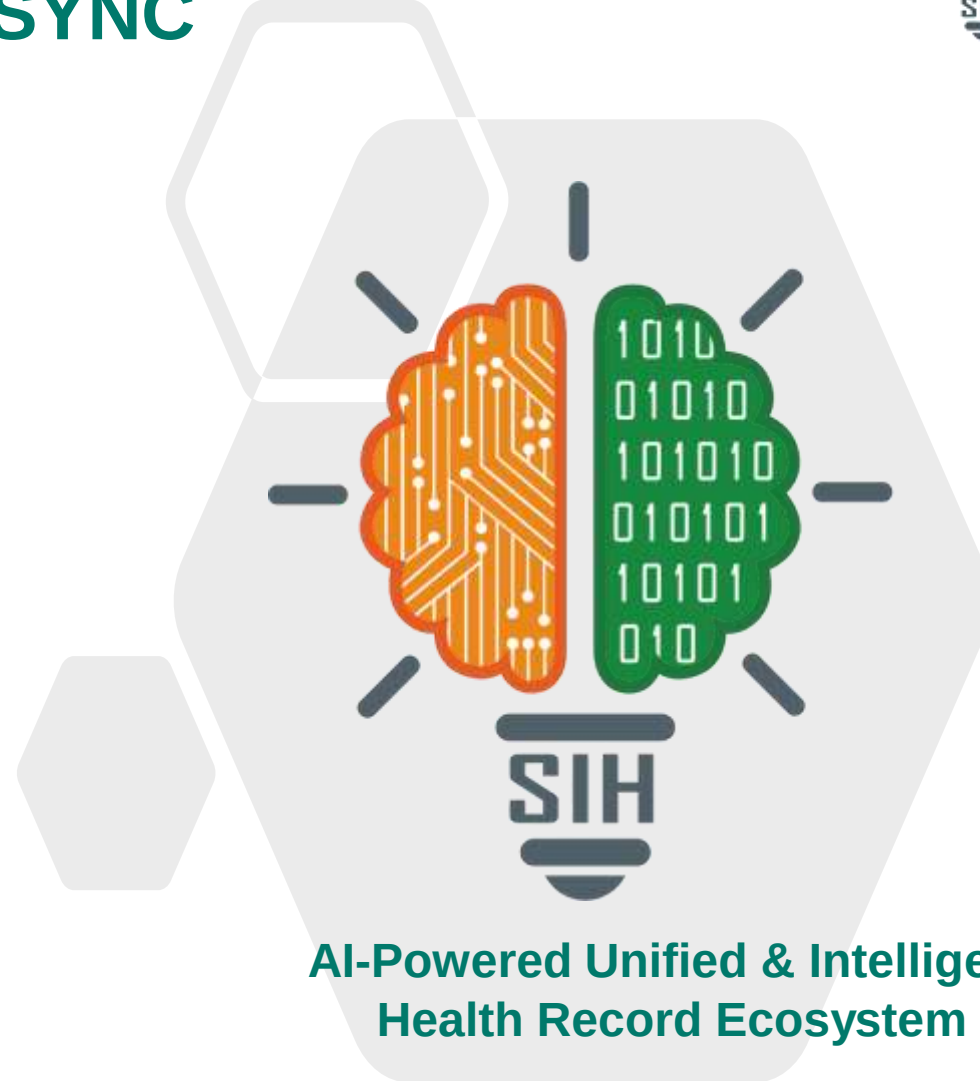


SMART INDIA HACKATHON 2026

CARESYNC



Problem Statement ID – SIH26198
Problem Statement Title – Student Innovation
Theme – MedTech / BioTech / HealthTech
PS Category – Software
Team ID – SIH361
Team Name – RECYK



**AI-Powered Unified & Intelligent
Health Record Ecosystem**

CARESYNC — PROPOSED SOLUTION

One patient. One intelligent health timeline. Better continuity of care.

THE PROBLEM

- Records are scattered across hospitals, labs and paper files.
- Doctors often lack a complete longitudinal view.
- Patients struggle to understand trends across old reports.

THE SOLUTION

- Patient-controlled unified health timeline.
- AI extracts key data from reports & prescriptions.
- Concise doctor summary + trend detection.
- Consent-based, time-limited record sharing.

WHAT MAKES IT UNIQUE

- Converts documents into structured clinical context.
- Highlights longitudinal changes—not isolated reports.
- Emergency QR exposes only essential authorized data.
- FHIR-ready interoperability layer.

UPLOAD / IMPORT



AI STRUCTURES



HEALTH TIMELINE



INSIGHTS



SHARE

TECHNICAL APPROACH



1. INPUT

Lab reports
Prescriptions
Discharge summaries

2. AI PIPELINE

OCR / parsing
Entity extraction
Normalization

3. HEALTH CORE

FHIR-ready records
Timeline
Consent metadata

4. INTELLIGENCE

Summary
Trend detection
Contextual alerts

5. ACCESS

Patient dashboard
Doctor view
Emergency QR

TECH STACK

Frontend

React / Next.js PWA

Backend

FastAPI / Node.js

Data

PostgreSQL + object storage

AI

Python NLP + OCR + LLM
summarization

Standards

HL7 FHIR-compatible model

Security

RBAC, encryption, audit logs, consent

FEASIBILITY AND VIABILITY



WHY IT IS FEASIBLE

- Web/PWA prototype needs no special hardware.
- OCR, NLP and FHIR libraries are mature.
- Modular APIs allow phased hospital/lab integration.
- MVP can start with patient-uploaded documents.

KEY RISKS

- Sensitive health-data privacy
- Inconsistent document formats
- AI extraction/summarization errors
- Integration with heterogeneous systems

MITIGATION

- Explicit consent + least-privilege access
- Encryption, audit trail and data minimization
- Confidence scores + human verification
- FHIR-based adapters and standard APIs

PHASED IMPLEMENTATION

MVP

Upload → extract → timeline

PHASE 2

Doctor sharing + emergency QR

PHASE 3

ABDM/FHIR integrations

SCALE

Hospitals, labs & population analytics

IMPACT AND BENEFITS



CareSync improves continuity of care by turning fragmented records into a portable, understandable health history.

PATIENTS

- Own and organize records
- Understand health trends
- Faster sharing during visits
- Better portability of history

DOCTORS

- Concise longitudinal summary
- Less time searching files
- Better context at point of care
- Reduced repeated history-taking

HEALTH SYSTEM

- Interoperable data flow
- Lower administrative friction
- Supports continuity across providers
- Foundation for responsible analytics

SOCIAL VALUE

- Useful for elderly/family care
- Multilingual accessibility
- Low-bandwidth PWA pathway
- Faster access to critical information

SUCCESS METRICS • Record retrieval time • Extraction accuracy • Clinician summary usefulness • Consent-controlled shares

RESEARCH AND REFERENCES



INDIA: ABDM / ABHA

ABDM Personal Health Records support linking, viewing and consent-based sharing of health records.
Reference: abdm.gov.in

INTEROPERABILITY: HL7 FHIR

FHIR provides standardized, web-friendly resources for interoperable healthcare data exchange.
Reference: hl7.org/fhir

WHO: CONTINUITY OF CARE

WHO highlights interoperable digital systems and portable health information as enablers of continuity of care.
Reference: who.int

PRIVACY & CONSENT

Health information requires medical confidentiality, appropriate consent and strong privacy safeguards.
Reference: WHO SMART Trust / ABDM

Design principle: CareSync complements national digital-health infrastructure; it does not replace clinical judgment. AI outputs are assistive, confidence-aware and reviewable.